

Chapter 1: Newton's Laws of Motion & Classical Dynamics

1.1 Introduction to Classical Mechanics & Inertia

The Historical Paradigm Shift: Aristotle to Galileo and Newton

For nearly two millennia, the Aristotelian view dominated natural philosophy: that a body requires a continuous, active application of force to sustain uniform motion, and that coming to rest is the natural state of all terrestrial objects.

Galileo Galilei challenged this doctrine through thought experiments involving smooth inclined planes. Galileo deduced that if friction could be eliminated entirely, a ball rolling down an incline and up another would attain the exact same height. If the second plane were flattened horizontally to infinity, the ball would roll forever at invariant velocity.

Sir Isaac Newton formalized this revolutionary insight in 1687 in the *Philosophiae Naturalis Principia Mathematica* as his First Law of Motion.

Key Formula / Law: *Newton's First Law (Law of Inertia): An object at rest remains at rest, and an object in motion continues in uniform motion along a straight line at constant speed, unless acted upon by a non-zero net external force ($\Sigma F = 0 \Rightarrow a = 0, v = \text{const}$).*

1.2 Newton's Second Law: Force, Mass, and Acceleration

Mathematical Formulation and Momentum

Newton did not originally state his second law simply as $F = ma$, but rather in terms of quantity of motion, which modern physics defines as linear momentum ($p = mv$).

The second law states that the rate of change of linear momentum of a body is directly proportional to the applied net external force and occurs along the line in which that force acts.

When mass remains invariant with respect to time ($dm/dt = 0$), differentiation yields the canonical equation:
 $F_{\text{net}} = d(mv)/dt = m(dv/dt) = m \cdot a$.

Key Formula / Law: *Newton's Second Law: $\Sigma F = m \cdot a$ | In SI Units: 1 Newton (N) = 1 kg · m/s² | Vector Form: $\Sigma F_x = m a_x, \Sigma F_y = m a_y$*

1.3 Newton's Third Law: Action-Reaction Interactions

Mutual Forces Between Two Distinct Bodies

Forces never occur in isolation; they are mutual interactions between pairs of bodies. Whenever body A exerts a force on body B ($F_{\text{A on B}}$), body B simultaneously exerts an equal magnitude, oppositely directed force back upon body A ($F_{\text{B on A}}$).

A critical conceptual pitfall among students is wondering why action and reaction forces do not cancel each other out to zero. The fundamental reason is that action and reaction forces act on two completely distinct physical bodies, never on the same object! Therefore, they can never balance each other in a single free-body diagram.

Key Formula / Law: Newton's Third Law: $F_{\{A \text{ on } B\}} = -F_{\{B \text{ on } A\}}$ (Equal magnitude, opposite collinear direction, acting on different bodies)

1.4 Free-Body Diagrams (FBD) and Friction

Systematic Problem-Solving Strategy

A Free-Body Diagram is an indispensable analytical schematic that isolates a single physical body and depicts all external forces acting upon it using vector arrows originating from its center of mass.

Frictional forces oppose relative tangential motion between surfaces in contact: 1. Static Friction (f_s): Resists the initiation of motion, satisfying $f_s \leq \mu_s \cdot N$, where μ_s is the coefficient of static friction and N is the normal contact force. 2. Kinetic Friction (f_k): Retards ongoing sliding motion, governed by $f_k = \mu_k \cdot N$ (where typically $\mu_k < \mu_s$).

Key Takeaways:

- Mass is an intrinsic quantitative measure of inertia; Weight is the gravitational force $W = mg$.
- An object in dynamic equilibrium (constant velocity) has zero net acceleration, hence $\Sigma F = 0$.
- Normal force is not automatically equal to mg ; it must be deduced from the equation of equilibrium along the axis perpendicular to the contact surface.
- Action-reaction pairs act on different objects and never cancel out on a single object's FBD.